

Software quality control techniques

- **Fault avoidance:** prevents errors before the system is released.
 - reviews, inspections, walkthroughs, development methodologies, testing, verification
- **Fault tolerance:** enables the system to recover from (some classes of) errors by itself.
 - rollbacks, redundancy, mirroring



Verification & Validation

- **Verification** is the process of evaluating a system or component to determine whether the products of a given development phase satisfy the conditions imposed at the start of that phase.
 - Are we building the product right?
 - Testing, Reviews
- **Validation** is the process of evaluating a system or component during or at the end of the development process to determine whether it satisfies specified requirements.
 - Are we building the right product?
 - Comparison against requirements

Terminology

- **Mistake** – a human action that produces an incorrect result.
- **Fault [or Defect]** – an incorrect step, process, or data definition in a program.
- **Failure** – the inability of a system or component to perform its required function within the specified performance requirement.
- **Error** – the difference between a computed, observed, or measured value or condition and the true, specified, or theoretically correct value or condition.



```
int factorial(int a){  
    if(a==1)  
        return 1;  
  
    return a* factorial(a-1);  
}
```

Mistake: Did not consider
numbers less than 1



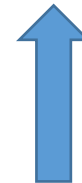
Exception or
error output
expected

Error



Overflow / crash

Failure



Lack of input
validation

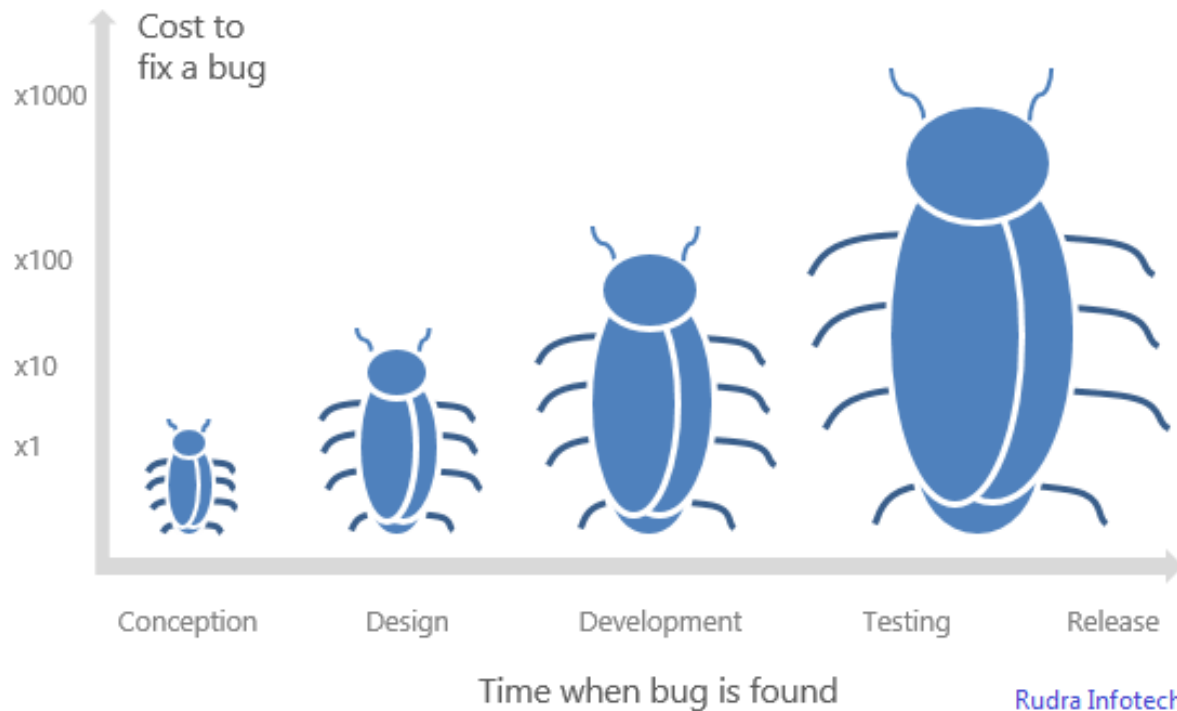
Defect: missing
validation

Software Testing

Software testing is the process of analyzing a software item to detect the differences between existing and required conditions (i.e., bugs) and to evaluate the features of the software item.

When should we test software?

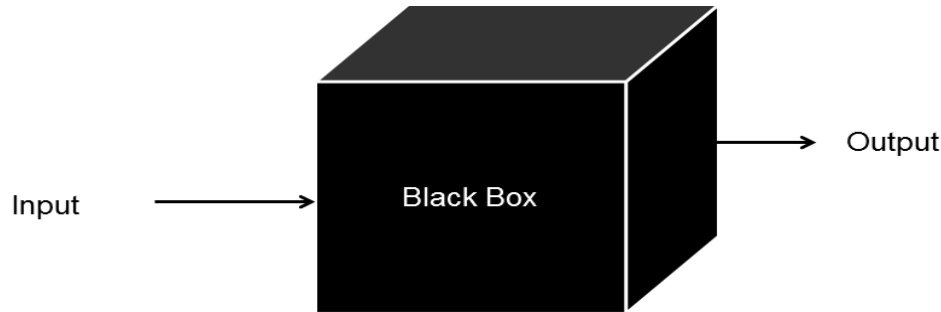
Answer: Throughout the whole development process



Testing goals

- Write test cases to cause failures.
- But, there is no way to guarantee that all faults have been detected.
- Work smart: write as few test cases as possible to cause failures; don't have more than one test cause the same failure

Testing model: Black box testing



- **Black box testing:** ignores the internal mechanism of a system or component and focuses solely on the outputs generated in response to selected inputs and execution conditions.
 - Interface visible, internals unknown
 - You know what it is supposed to do, you design tests that make it do what you think that it should do
 - From the outside, you are testing its functionality against the specs
 - For software this is testing the interface
 - What is input to the system?
 - What you can do from the outside to change the system?
 - What is output from the system?
 - Tests the functionality of the system by observing its external behavior
 - No knowledge of how it goes about meeting the goals

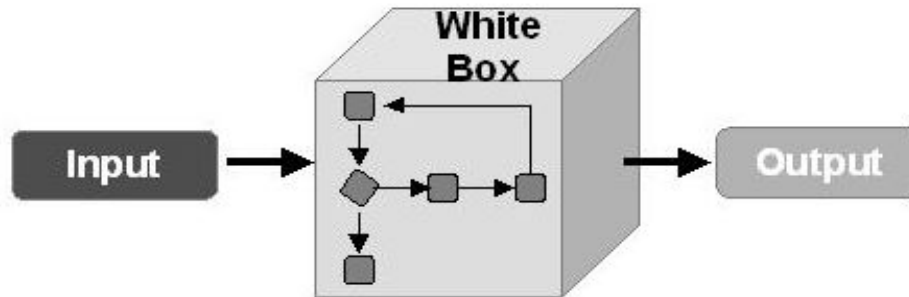
Advantages of Black box testing

- Process is not influenced by component being tested
 - Assumptions embodied in code not propagated to test data.
- Robust with respect to changes in implementation
 - Test data need not be changed when code is changed
- Allows for independent testers
 - Testers need not be familiar with code
- Useful for validation (are we building the right software)

Disadvantages of Black box testing

- It will miss bugs in the implementation that are not covered by the specification
 - Control-flow details
 - Performance optimizations
 - Alternate algorithms for different cases

Testing model: White box testing



- **White box testing** takes into account the internal mechanisms of a system or component.
 - Given knowledge of the internal workings, you thoroughly test what is happening on the inside
 - Close examination of procedural level of detail
 - Logical paths through code are tested
 - Conditionals
 - Loops
 - Branches
 - Status is examined in terms of expected values

Advantages of White box testing

- Finds an important class of boundaries.
- Yields useful test cases.
- Very useful to examine and test important data structures
- Useful for verification (are we building the software right)

Disadvantages of White box testing

- Tests may have the same bugs as implementation!
- Impossible to thoroughly exercise all paths
 - Exhaustive testing grows without bound

Types of testing

- Six types of testing
 - Unit testing
 - Integration testing
 - Non-functional testing
 - Acceptance testing
 - Regression testing
 - Beta testing

Unit testing

- Testing of individual hardware or software units or groups of related units
- Done by programmer(s)
- Generally all white box
- Verify that code does what it is intended to do at a very low structural level
- Automation desirable for repeatability



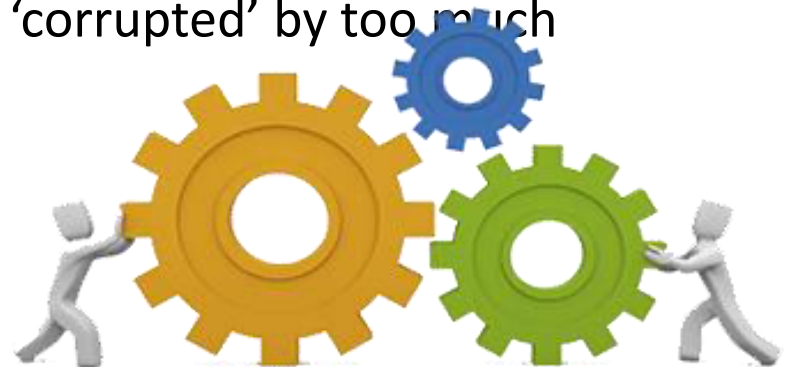
Integration testing

- Testing in which software components, hardware components, or both are combined and tested to evaluate the interaction between them
- Done by programmer as they integrate their code into code base
- Verifies that units work together when they are integrated into a larger code base
 - Just because the components work individually, that does not mean that they all work together when integrated
- Generally white box, maybe some black box
- Automation desirable for repeatability



Non-Functional testing

- Testing conducted on a complete, integrated system to evaluate the system compliance with its specified requirements
- *Stress testing*: evaluating a system beyond the limits of its specification
- *Performance/load testing*: evaluating the compliance of a system with specified performance requirements.
- *Usability testing*: evaluating the extend to which a user can learn to operation, prepare inputs for, and interpret outputs of a system.
- It is recommended that this be done by external test group
- Mostly black box so that testing is not 'corrupted' by too much knowledge



Acceptance testing

- Formal testing conducted to determine whether or not a system satisfies its acceptance criteria (the criteria the system must satisfy to be accepted by a customer) and to enable the customer to determine whether or not to accept the system
- Generally done by customer/customer representative in their environment through the GUI
- Definitely black box



Regression testing

- Regression testing is selective retesting of a system or component to verify that modifications have not caused unintended effects and that the system or component still complies with its specified requirements
- Subset of the original set of test cases.
- Core group of tests re-run often after any significant changes
 - Choose a representative sample of tests that exercise all the existing functionalities
 - Chose additional test cases that are most likely to be affected by the change
- *Smoke test*: a subset of the regression test cases that establish that the system is stable and all major functionality is present and works under “normal” conditions

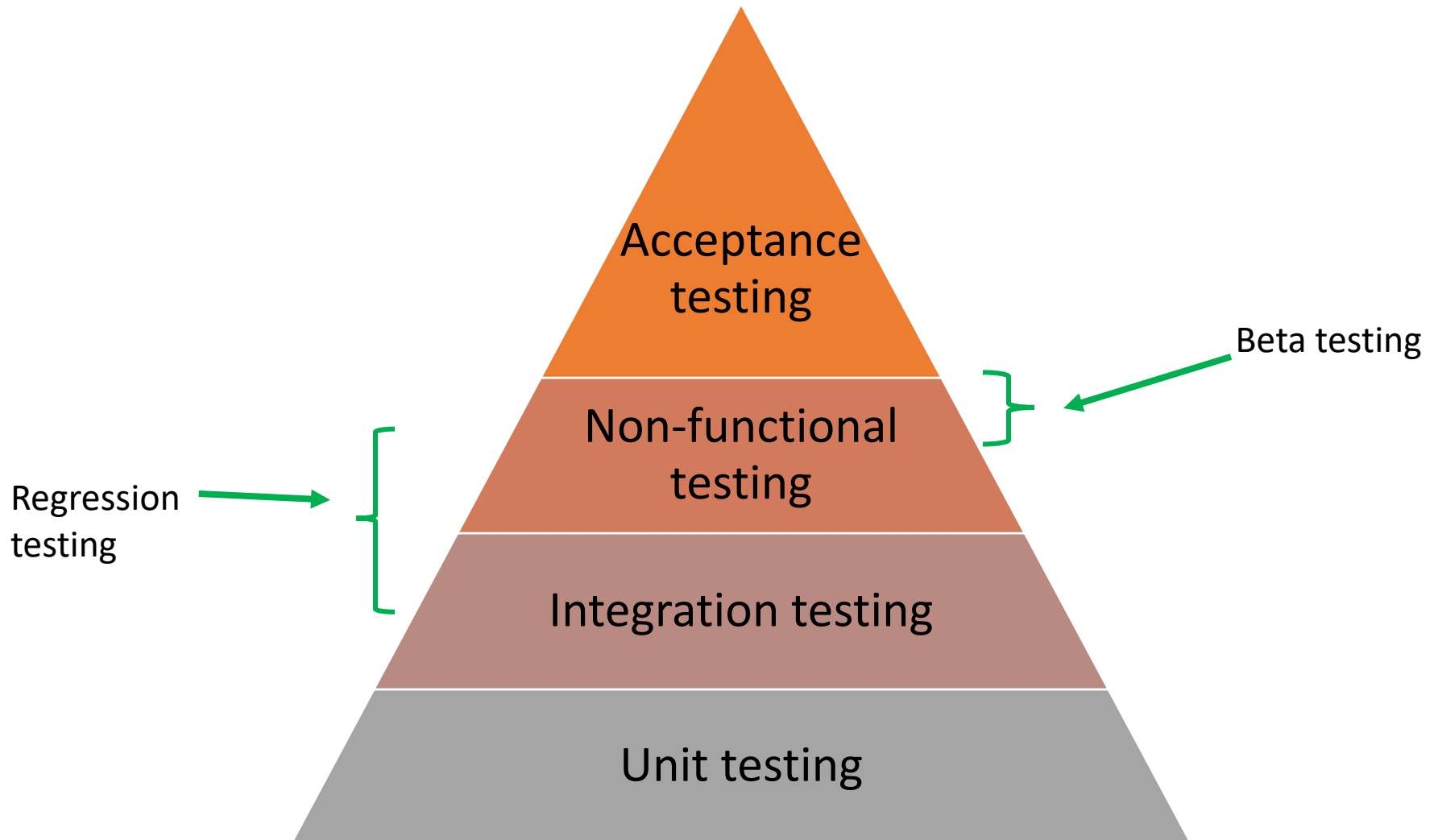


Beta testing

- Organization can offer an advance partial or full version of a software package free to one or more potential users.
- Users use the software as they wish, with the understanding that they will report any errors revealed during usage back to the organization.
- Advantages
 - Identification of unexpected errors
 - Low costs
 - Wider population / environment
- Disadvantages
 - Lack of systematic testing
 - Low quality error reports



Testing Hierarchy



Test plan

- Document describing the scope, approach, resources, and schedule of intended testing activities
- A test plan identifies
 - test items
 - features to be tested
 - Testing tasks
 - who will do the testing
 - any risks requiring contingency plans
- Test throughout the development cycle
- Write the test plan early in the development cycle
- If you wait until the end of the cycle, you might be in a very chaotic, hurried environment